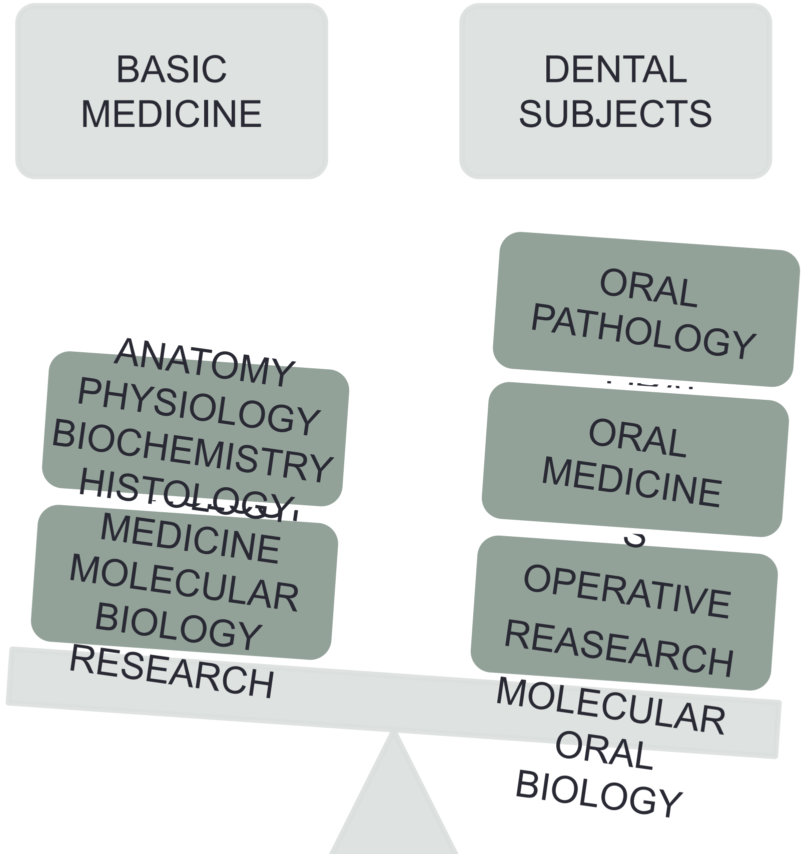
INTRODUCTION TO ORAL BIOLOGY COURSE

- The course of oral biology deals with the origin, growth and development, structure and functions of oral tissues.

- Oral biology overlaps the basic medical sciences and clinical dental sciences.

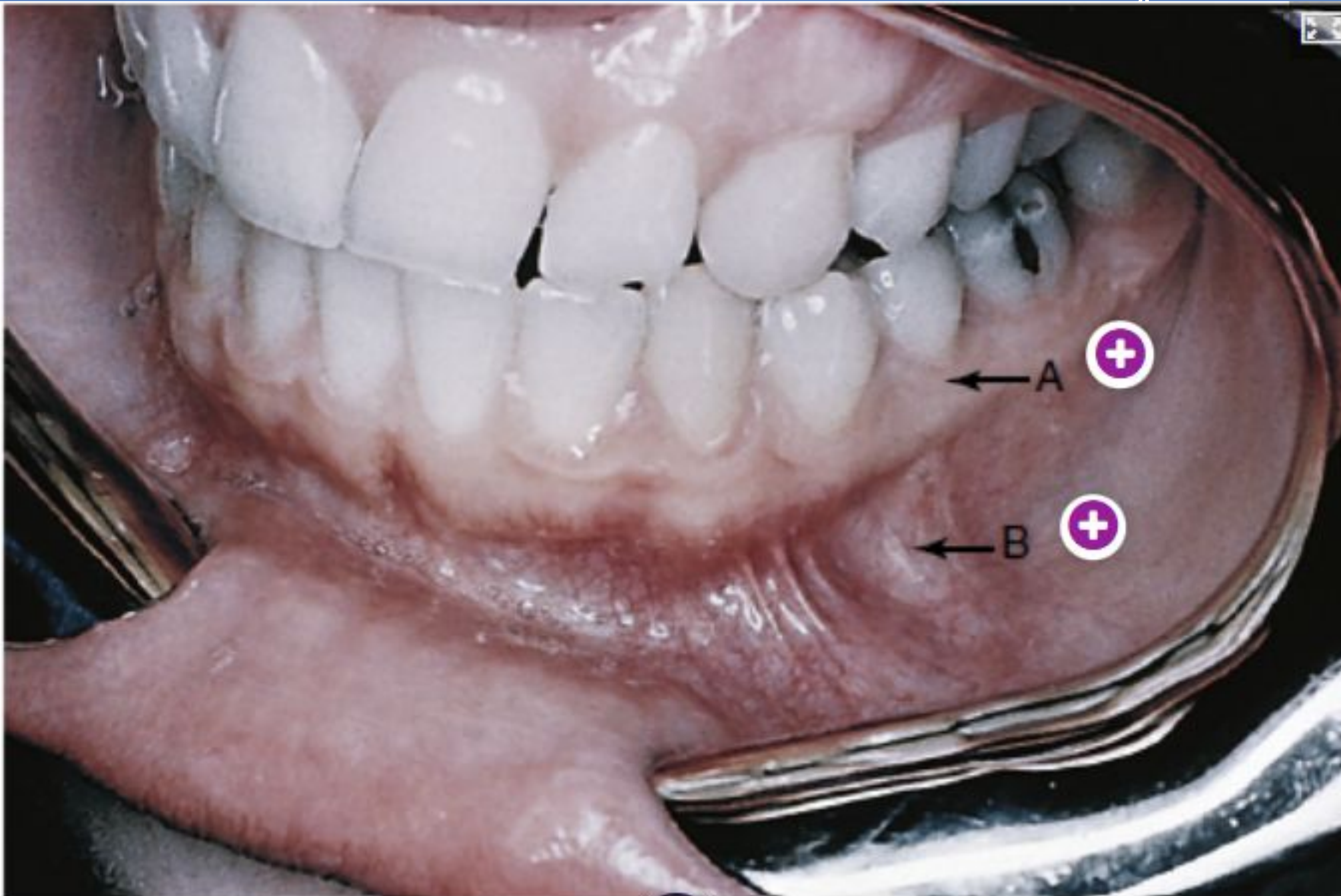
INTRODUCTION TO SUBJECT OF ORAL BIOLOGY



INTRODUCTION TO ORAL BIOLOGY COURSE

ORAL BIOLOGY

- ORAL ANATOMY , HISTOLOGY, PHYSIOLOGY
- TOOTH MORPHOLOGY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Oral histology is the scientific study of the microscopic structure and function of the tissues in the mouth, including the hard tissues such as enamel and dentin, as well as the soft tissues like the gums and oral mucosa.
This field is crucial for understanding how oral health issues develop and are treated, providing insights into conditions like dental caries and periodontal disease.
By studying oral histology, students can gain a deeper comprehension of oral anatomy and pathology, which is foundational for careers in dentistry and related fields.

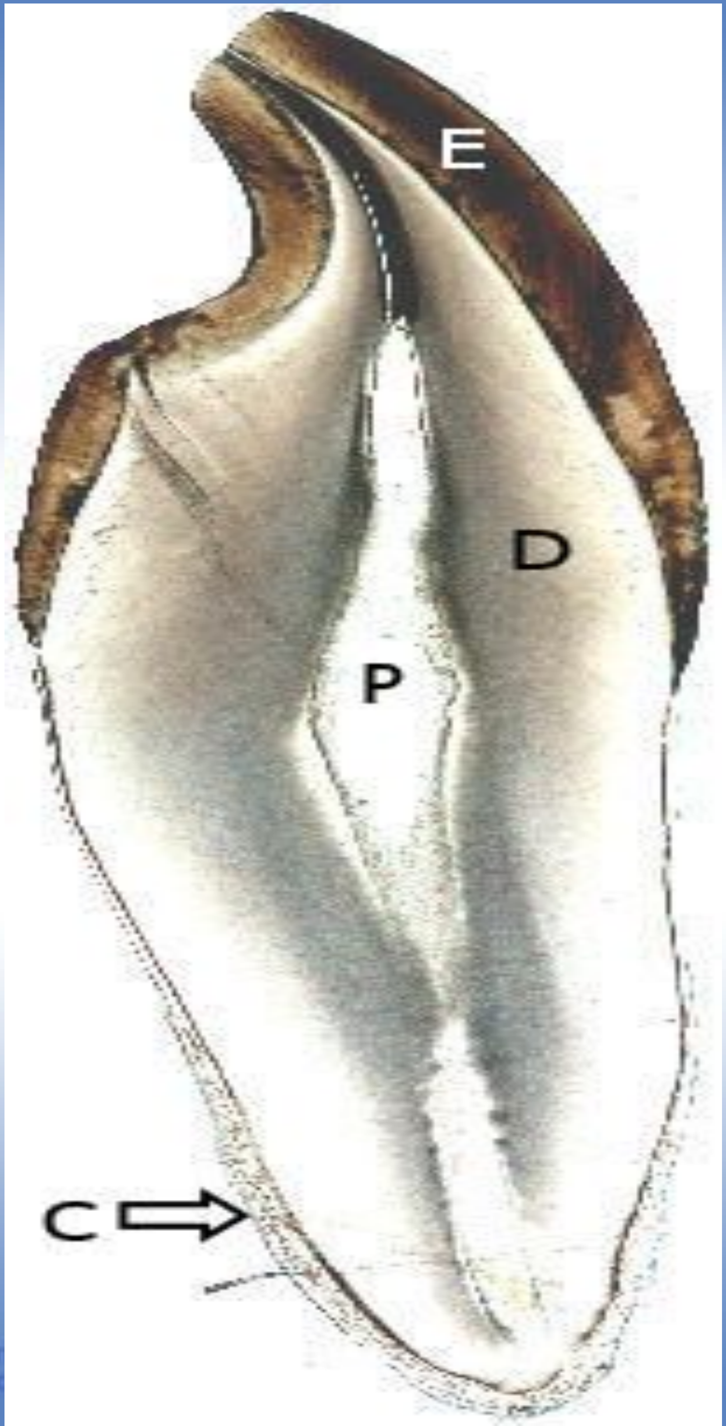
Oral histology

- The science dealing with the microscopic identification of cells and tissue in the oral cavity.

- The structure of organ tissues, including the composition of cells and their organization into various body tissues.

Light microscope

Electron microscope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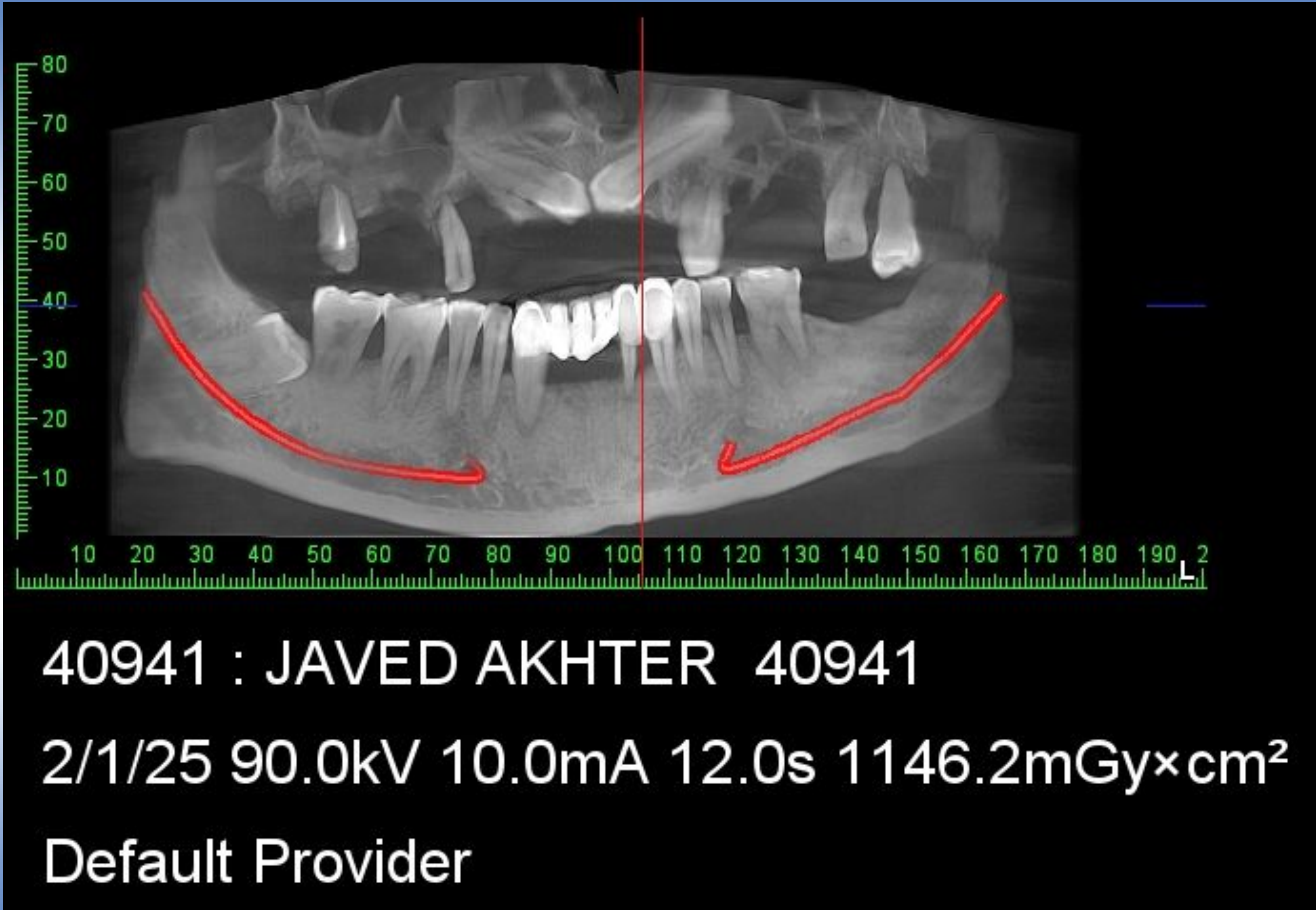
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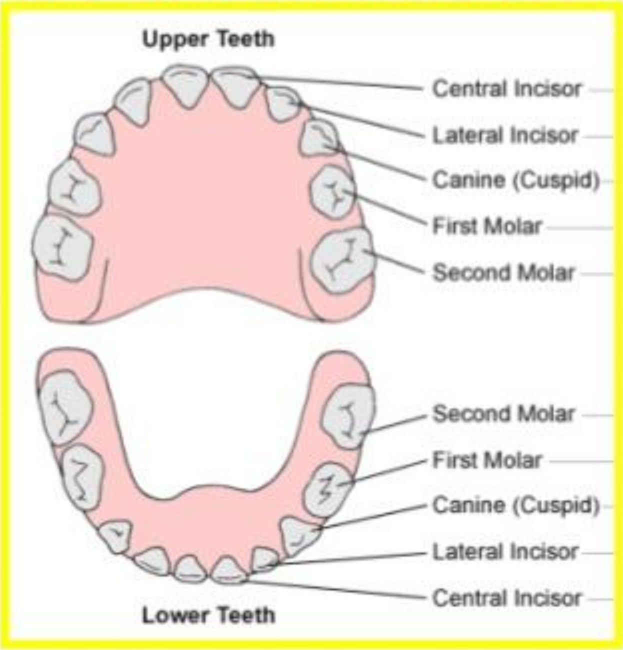
Human dentitions



1. What role does tooth structure play in the overall function of the digestive system?
2. How does understanding tooth morphology contribute to advancements in dental health?
3. In what ways can the study of histology inform us about oral diseases?
4. How does variation in tooth morphology among different species reflect their diet and lifestyle?
5. What are the implications of neglecting oral health on systemic health?

Dental Anatomy and Physiology

After viewing this lecture, attendees should be able to:

- Identify the major structures of the dental anatomy
- Discuss the primary characteristics of enamel, dentin, cementum, and dental pulp
- Describe the biologic functions that take place within the oral cavity

Dental Anatomy and Physiology

Definition (teeth): There are two definitions

- Primary (deciduous)
- Secondary (permanent)

Dental Anatomy and Physiology

Elements

A tooth is made up of three elements:

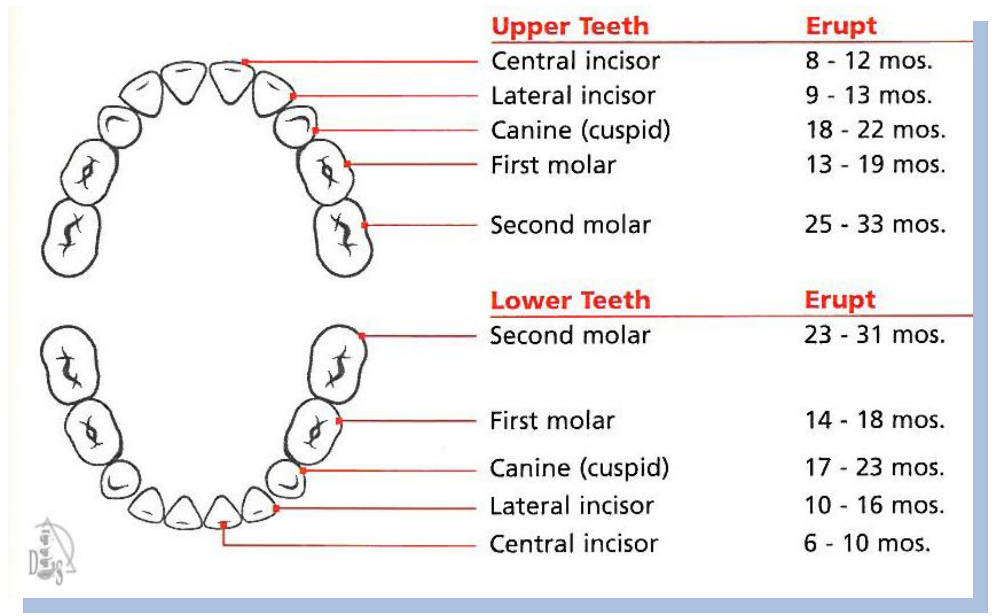
- Water
- Organic materials
- Inorganic materials

Dental Anatomy and Physiology

Dentition (teeth): There are two dentitions

Primary (deciduous)

- Consist of 20 teeth
- Begin to form during the first trimester of pregnancy
- Typically begin erupting around 6 months
- Most children have a complete primary dentition by 3 years of age



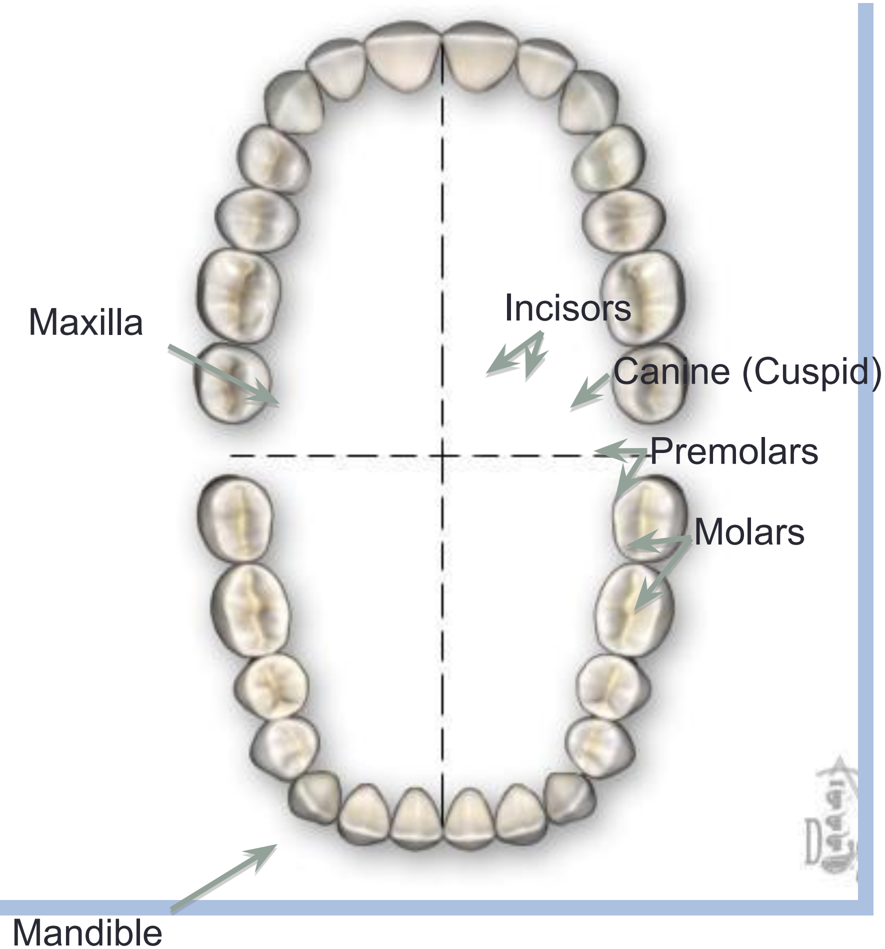
1. Oral Health for Children: Patient Education Insert. *Compend Cont Educ Dent*.

Dental Anatomy and Physiology

Dentition (teeth):
There are two dentitions

Secondary (permanent)

- Consist of 32 teeth in most cases
- Begin to erupt around 6 years of age
- Most permanent teeth have erupted by age 12
- Third molars (wisdom teeth) are the exception; often do not appear until late teens or early 20s

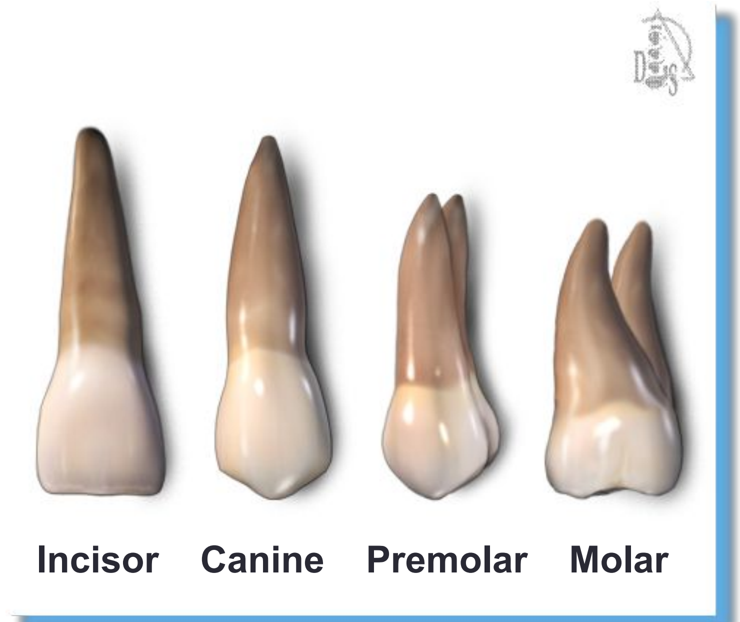


Dental Anatomy and Physiology

Identifying Teeth

Classification of Teeth:

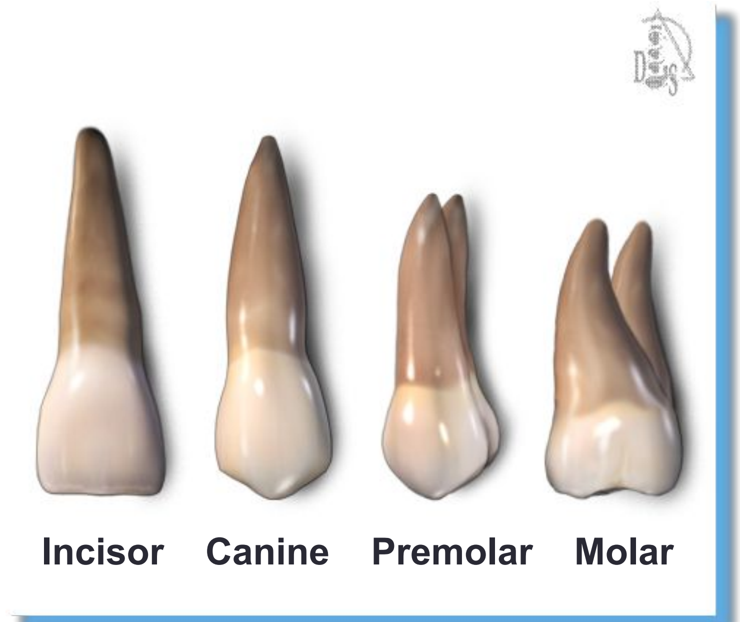
- Incisors (central and lateral)
- Canines (cuspids)
- Premolars (bicuspids)
- Molars



Dental Anatomy and Physiology

Identifying Teeth[2]

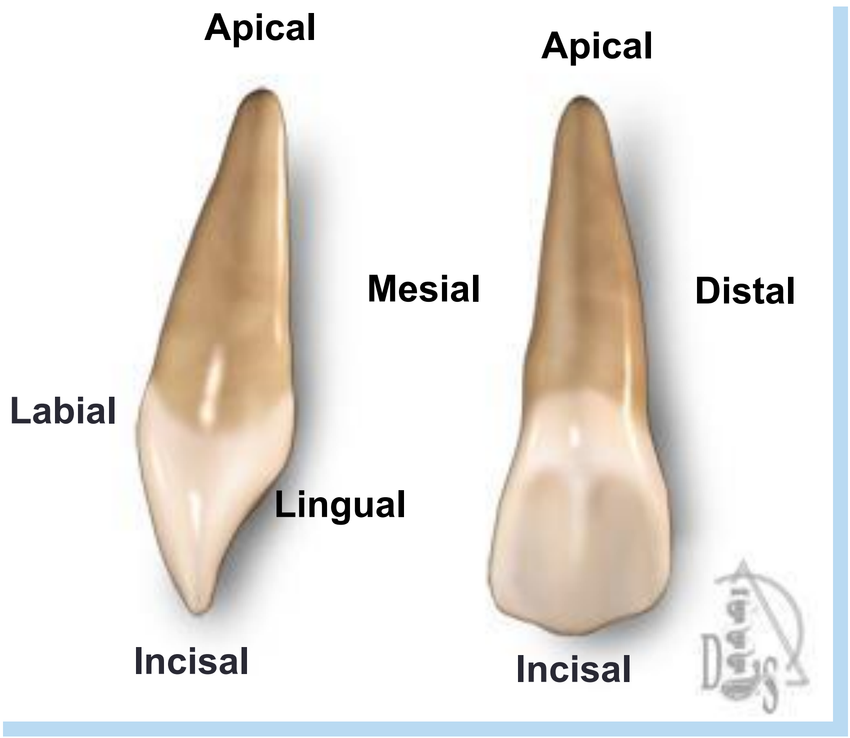
- **Incisors** function as cutting or shearing instruments for food.
- **Canines** possess the longest roots of all teeth and are located at the corners of the dental arch.
- **Premolars** act like the canines in the tearing of food and are similar to molars in the grinding of food.
- **Molars** are located nearest the temporomandibular joint (TMJ), which serves as the fulcrum during function.



Dental Anatomy and Physiology

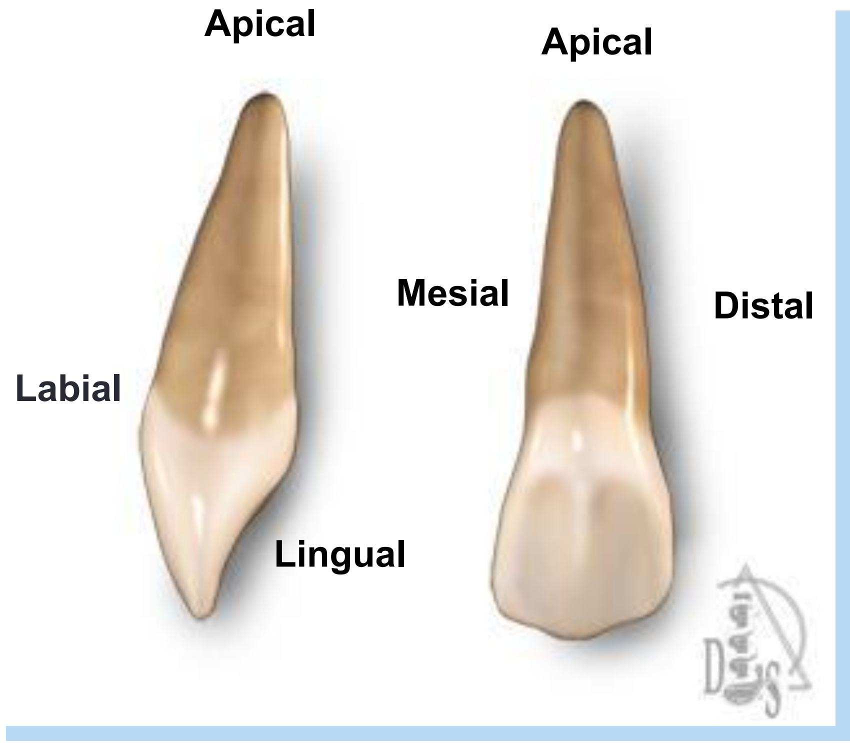
Teeth: Identification
Tooth Surfaces

- Apical
- Labial
- Lingual
- Distal
- Mesial
- Incisal



Dental Anatomy and Physiology

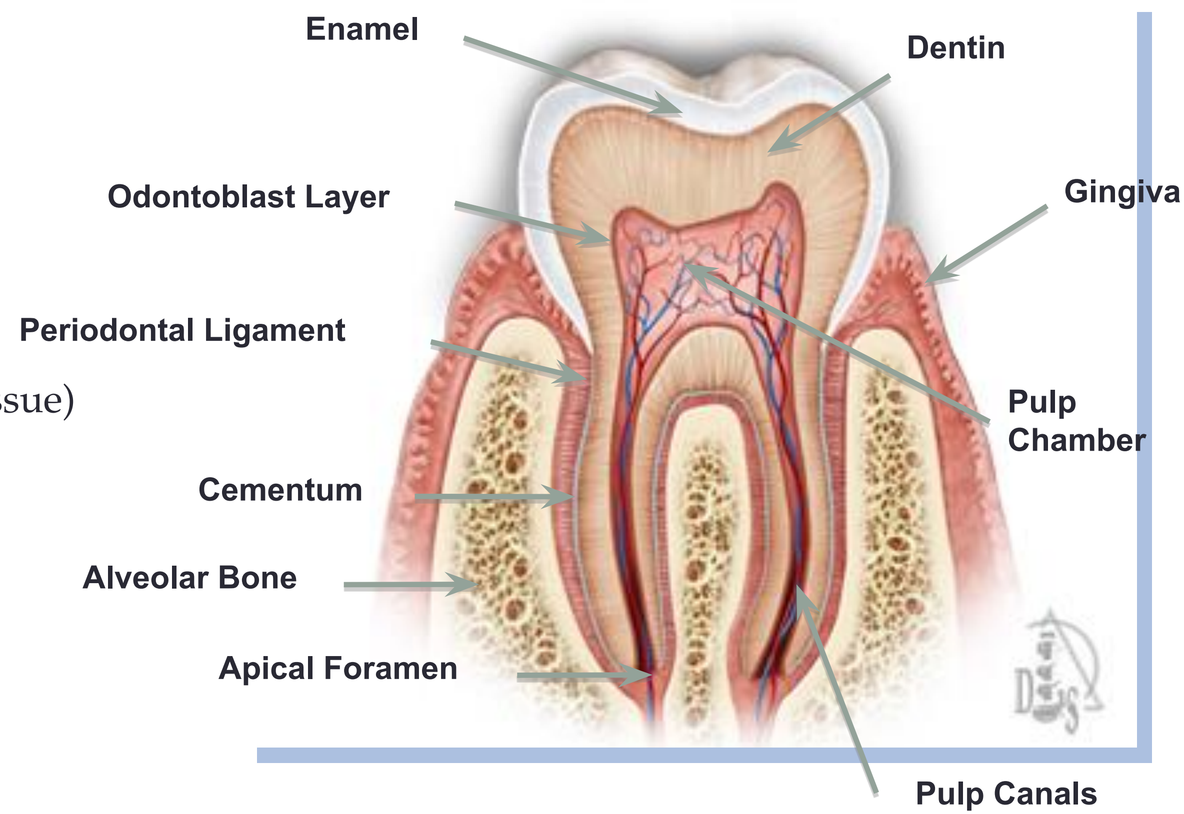
- **Apical**: Pertaining to the apex or root of the tooth
- **Labial**: Pertaining to the lip; describes the front surface of anterior teeth
- **Lingual**: Pertaining to the tongue; describes the back (interior) surface of all teeth
- **Distal**: The surface of the tooth that is away from the median line
- **Mesial**: The surface of the tooth that is toward the median line



Dental Anatomy and Physiology

The Dental Tissues:

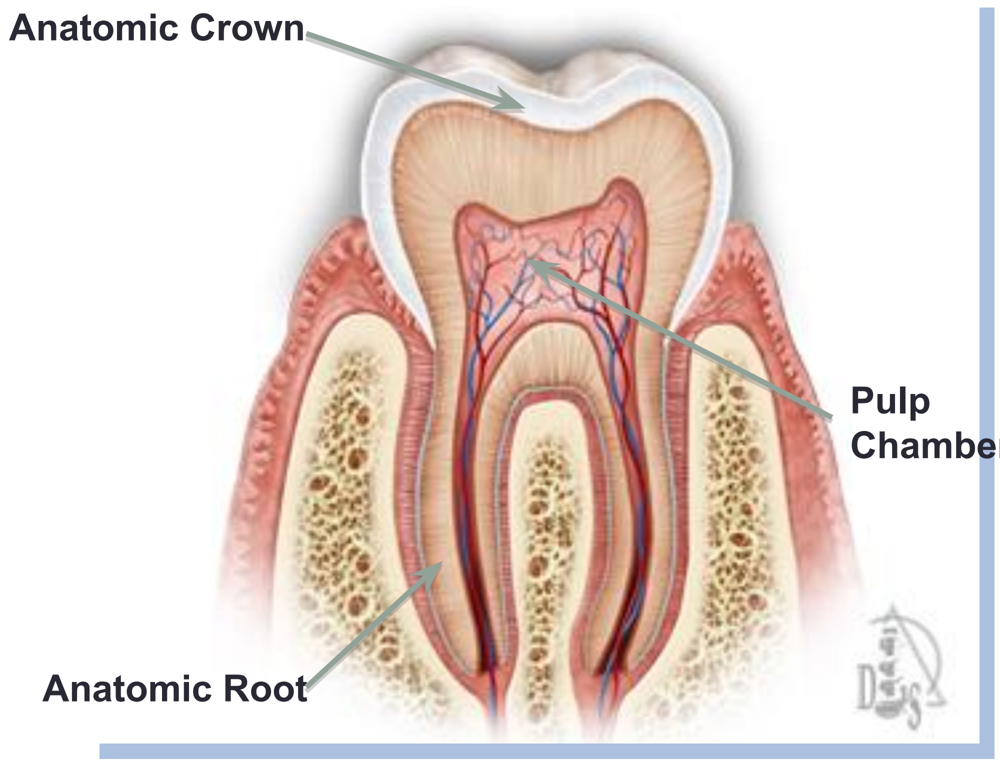
- Enamel (hard tissue)
- Dentin (hard tissue)
- Odontoblast Layer
- Pulp Chamber (soft tissue)
- Gingiva (soft tissue)
- Periodontal Ligament (soft tissue)
- Cementum (hard tissue)
- Alveolar Bone (hard tissue)
- Pulp Canals
- Apical Foramen



Dental Anatomy and Physiology

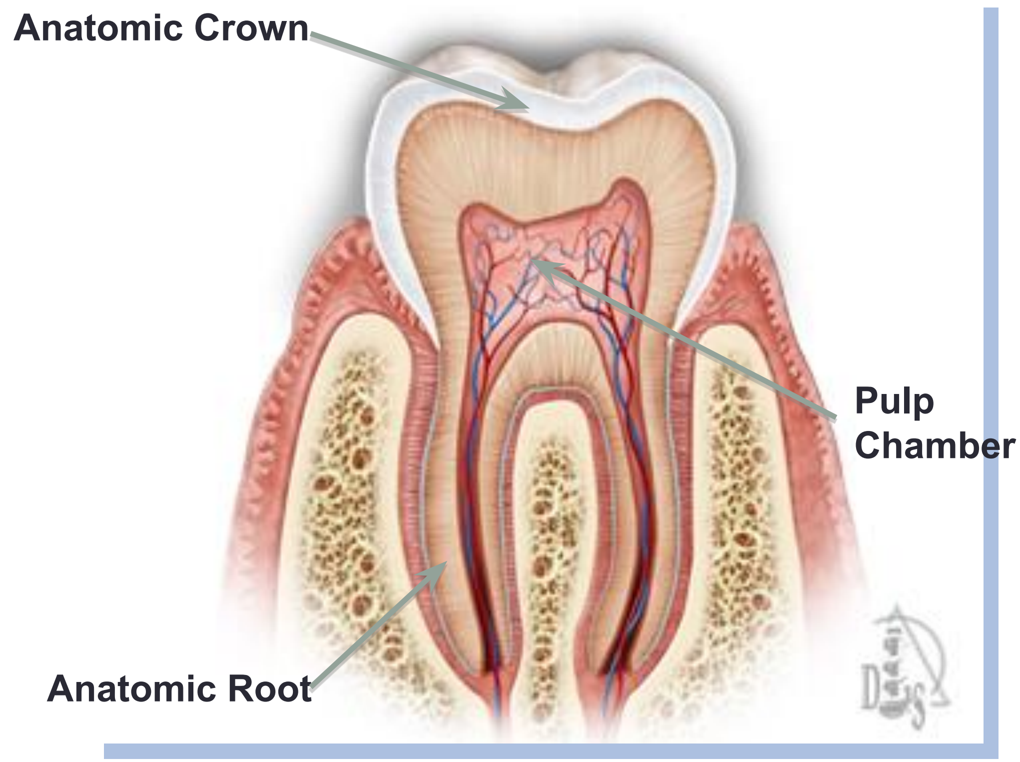
The 3 parts of a tooth:

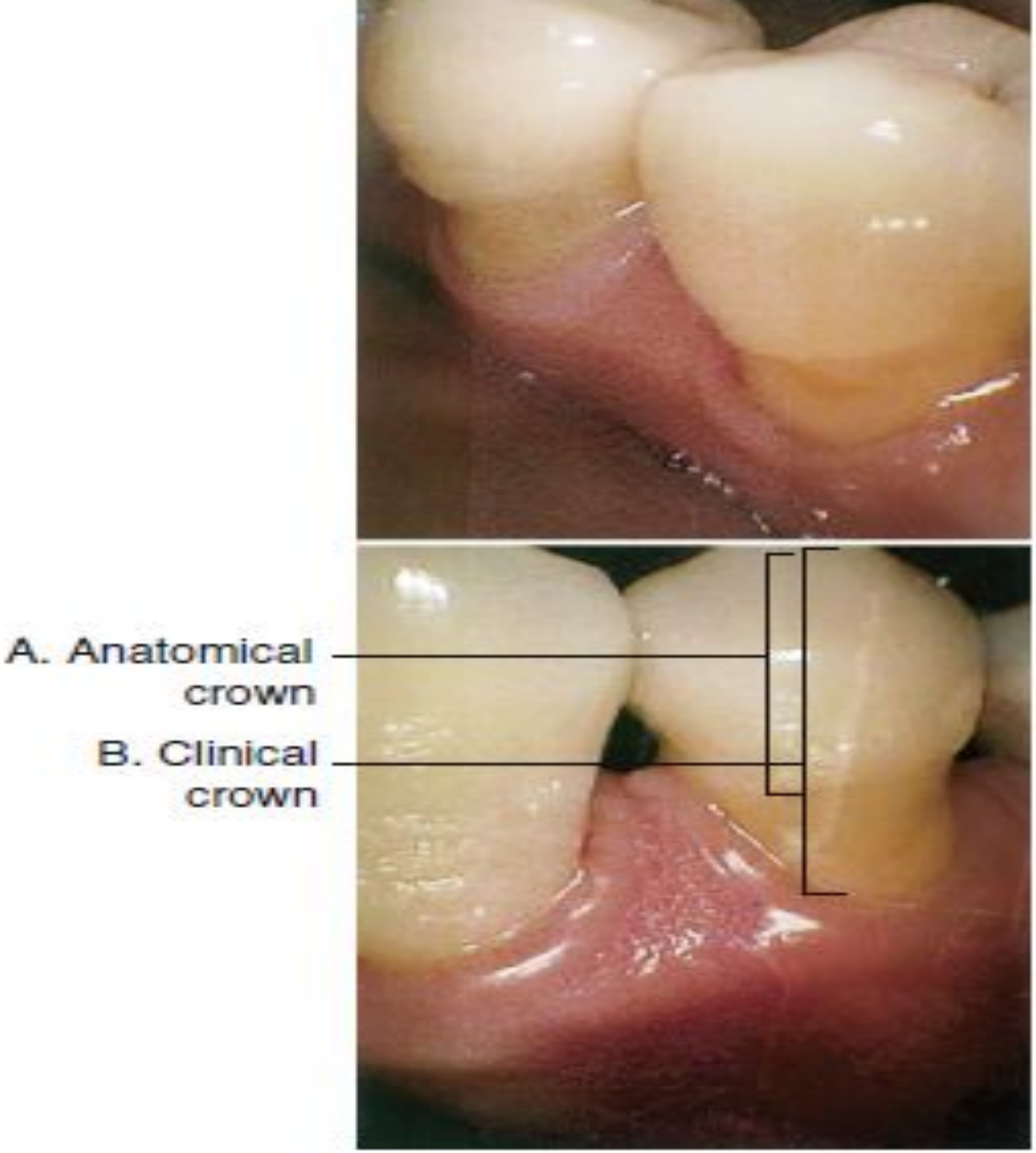
- Anatomic Crown
- Anatomic Root
- Pulp Chamber



Dental Anatomy and Physiology

- **The anatomic crown** is the portion of the tooth covered by enamel.
- **The anatomic root** is the lower two thirds of a tooth.
- **The pulp chamber** houses the dental pulp, an organ of myelinated and unmyelinated nerves, arteries, veins, lymph channels, connective tissue cells, and various other cells.



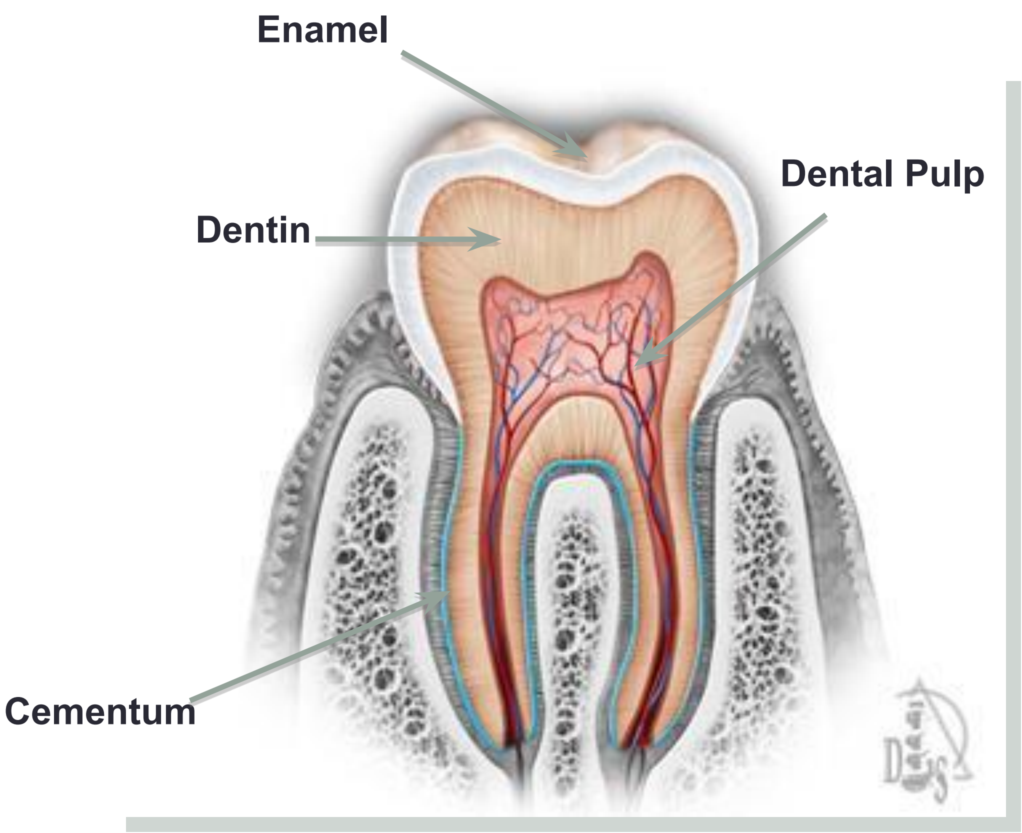


An anatomical crown and a clinical

Dental Anatomy and Physiology

The 4 main dental tissues:

- Enamel
- Dentin
- Cementum
- Dental Pulp



Dental Anatomy and Physiology

Dental Tissues—Enamel[2]

- Structure
  - Highly calcified and hardest tissue in the body
  - Crystalline in nature
  - Enamel rods
- Insensitive—no nerves
- Acid-soluble—will demineralize at a pH of 5.5 and lower
- Cannot be renewed
- Darkens with age as enamel is lost
- Fluoride and saliva can help with remineralization

Dental Anatomy and Physiology

Dental Tissues—Enamel[2]

- Enamel can be lost by:[3,4]
  - Physical mechanism
    - Abrasion (mechanical wear)
    - Attrition (tooth-to-tooth contact)
- Abfraction (lesions)
  - Chemical dissolution
    - Erosion by extrinsic acids (from diet)
    - Erosion by intrinsic acids (from the oral cavity/digestive tract)
- Multifactorial etiology
  - Combination of physical and chemical factors

: Having to do with the tooth's root.

: Part of tooth that shows in the mouth.

: Having to do with the tooth's crown.

: Part of tooth that shows in the mouth.

: Having to do with the tooth's apex.

: Part of the tooth where enamel and cementum meet

Radicular | Cementoenamel Junction | Apical | Anatomic Crown | Clinical Crown | Coronal

Dental Anatomy and Physiology

Dental Tissues—Dentin[2]

- Softer than enamel
- Susceptible to tooth wear (physical or chemical)
- Does not have a nerve supply but can be sensitive
- Is produced throughout life
- Three classifications
  - Primary
  - Secondary
  - Tertiary
- Will demineralize at a pH of 6.5 and lower

Dental Anatomy and Physiology

Dental Tissues—Dentin[2]

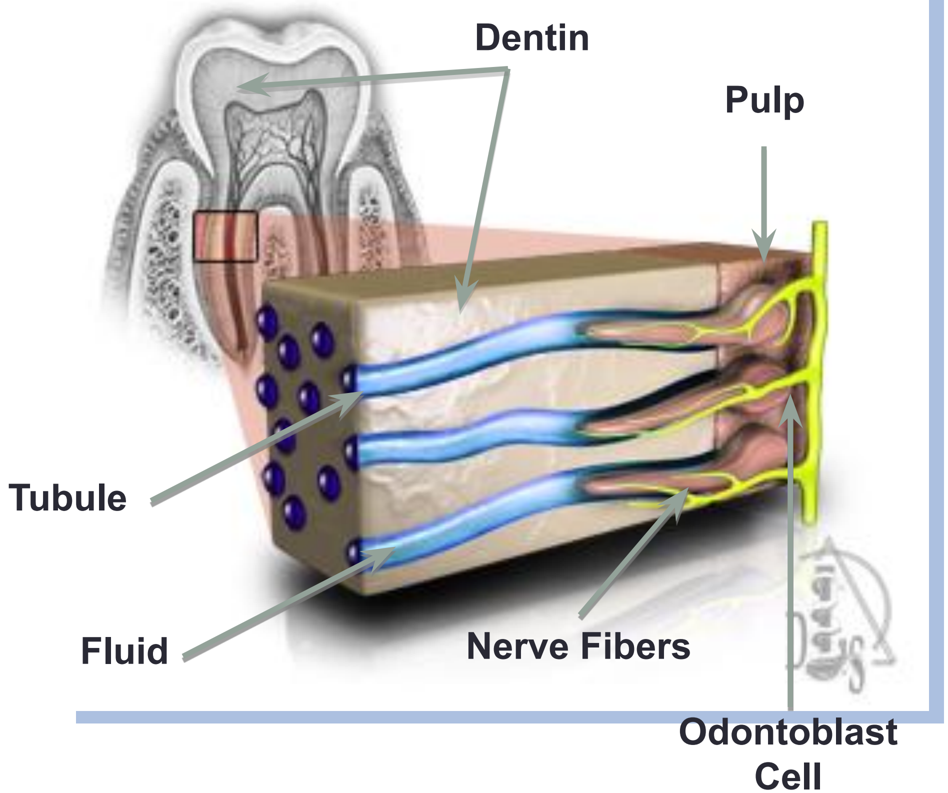
Three classifications:

- **Primary dentin** forms the initial shape of the tooth.
- **Secondary dentin** is deposited after the formation of the primary dentin on all internal aspects of the pulp cavity.
- **Tertiary dentin**, or “reparative dentin” is formed by replacement odontoblasts in response to moderate-level irritants such as attrition, abrasion, erosion, trauma, moderate-rate dental caries, and some operative procedures.

Dental Anatomy and Physiology

Dental Tissues—Dentin (Tubules)[2]

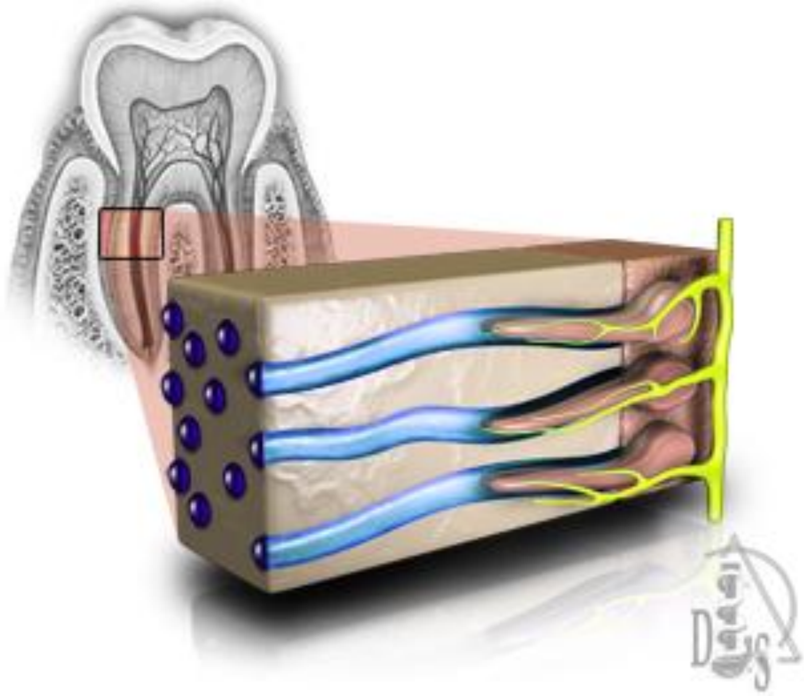
- **Dentinal tubules** connect the **dentin** and the **pulp** (innermost part of the tooth, circumscribed by the dentin and lined with a layer of **odontoblast cells**)
- The tubules run parallel to each other in an S-shape course
- Tubules contain **fluid** and **nerve fibers**
- External stimuli cause movement of the dentinal fluid, a hydrodynamic movement, which can result in short, sharp pain episodes



Dental Anatomy and Physiology

Dental Tissues—Dentin (Tubules)[2]

- Presence of tubules renders dentin permeable to fluoride
- Number of tubules per unit area varies depending on the location because of the decreasing area of the dentin surfaces in the pulpal direction

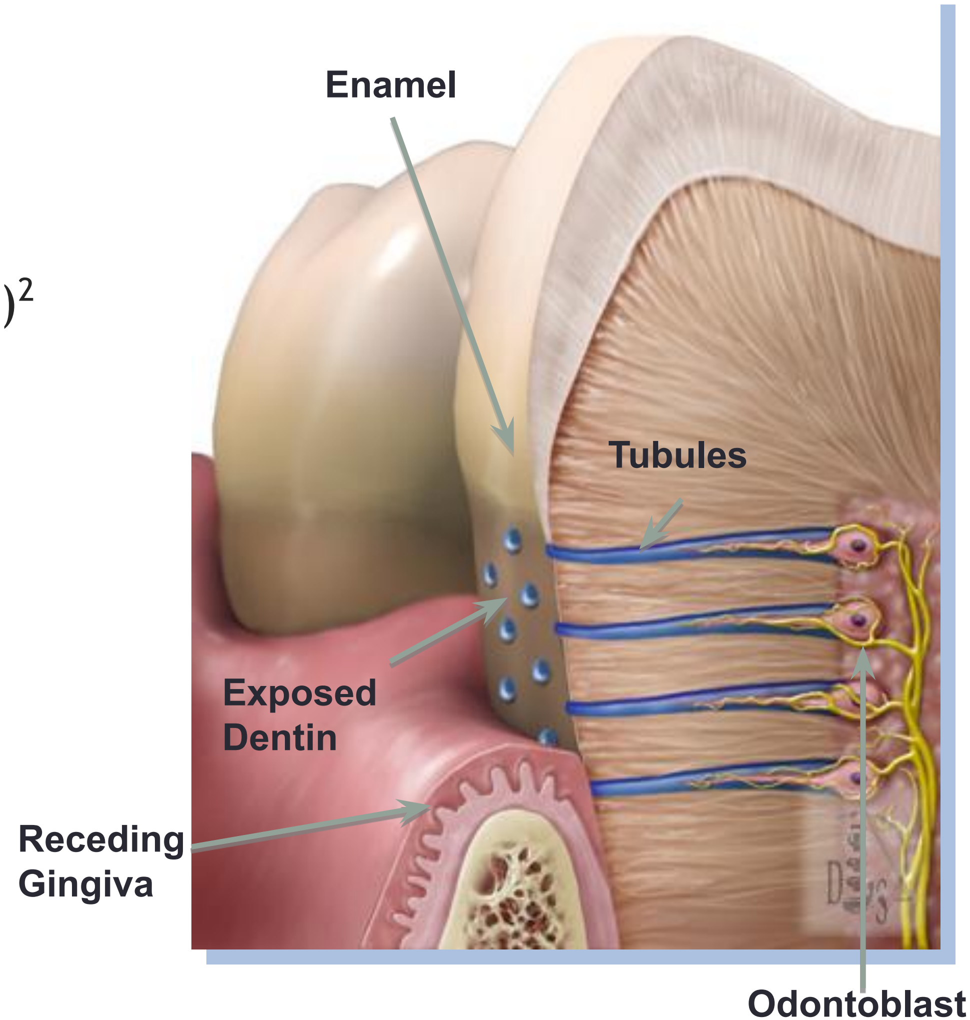


Dental Anatomy and Physiology

Dental Tissues—Dentin (Tubules)[2]

Association between erosion and dentin hypersensitivity[3]

- Open/patent tubules
  - Greater in number
  - Larger in diameter
- Removal of smear layer
- Erosion/tooth wear



Dental Anatomy and Phy

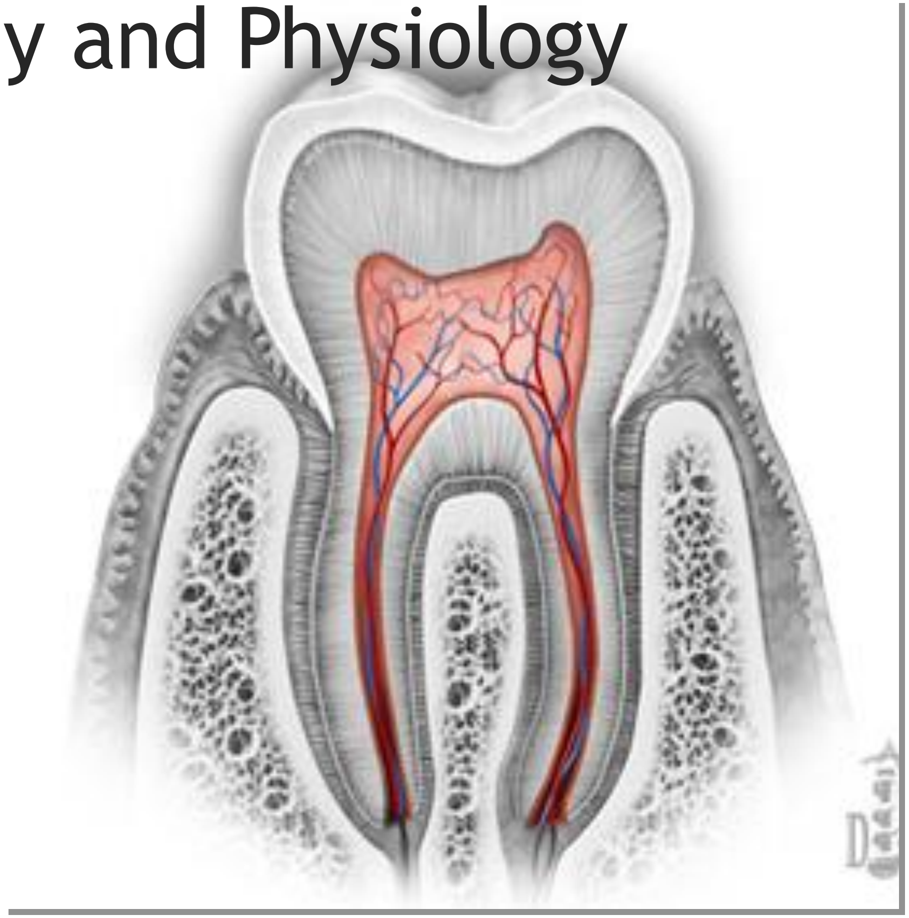
Dental Tissue—Cementum[2]

- Thin layer of mineralized tissue covering the dentin
- Softer than enamel and dentin
- Anchors the tooth to the alveolar bone along with the periodontal ligament
- Not sensitive

Dental Anatomy and Physiology

Dental Tissue—Dental Pulp[2]

- Innermost part of the tooth
- A soft tissue rich with blood vessels and nerves
- Responsible for nourishing the tooth
- The pulp in the crown of the tooth is known as the coronal pulp
- Pulp canals traverse the root of the tooth
- Typically sensitive to extreme thermal stimulation (hot or cold)



Dental Anatomy and Physiology

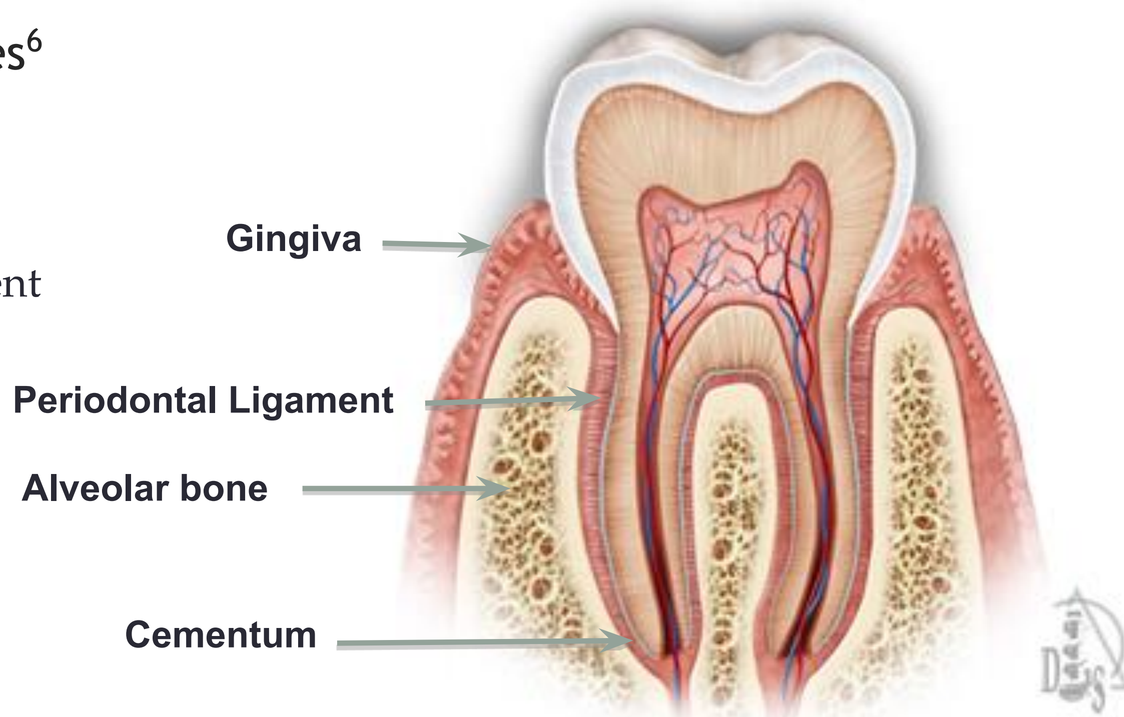
Dental Tissue—Dental Pulp[2,5]

- **Pulpitis** is inflammation or infection of the dental pulp, causing extreme sensitivity and/or pain.
- Pain is derived as a result of the hydrodynamic stimuli activating mechanoreceptors in the nerve fibers of the superficial pulp (A-beta, A-delta, C-fibers).
- Hydrodynamic stimuli include: thermal (hot and cold); tactile; evaporative; and osmotic
- These stimuli generate inward or outward movement of the fluid in the tubules and activate the nerve fibers.
- A-beta and A-delta fibers are responsible for sharp pain of short duration
- C-fibers are responsible for dull, throbbing pain of long duration
- Pulpitis may be reversible (treated with restorative procedures) or irreversible (necessitating root canal).
- Untreated pulpitis can lead to pulpal necrosis necessitating root canal or extraction.

Dental Anatomy and Physiology

Periodontal Tissues[6]

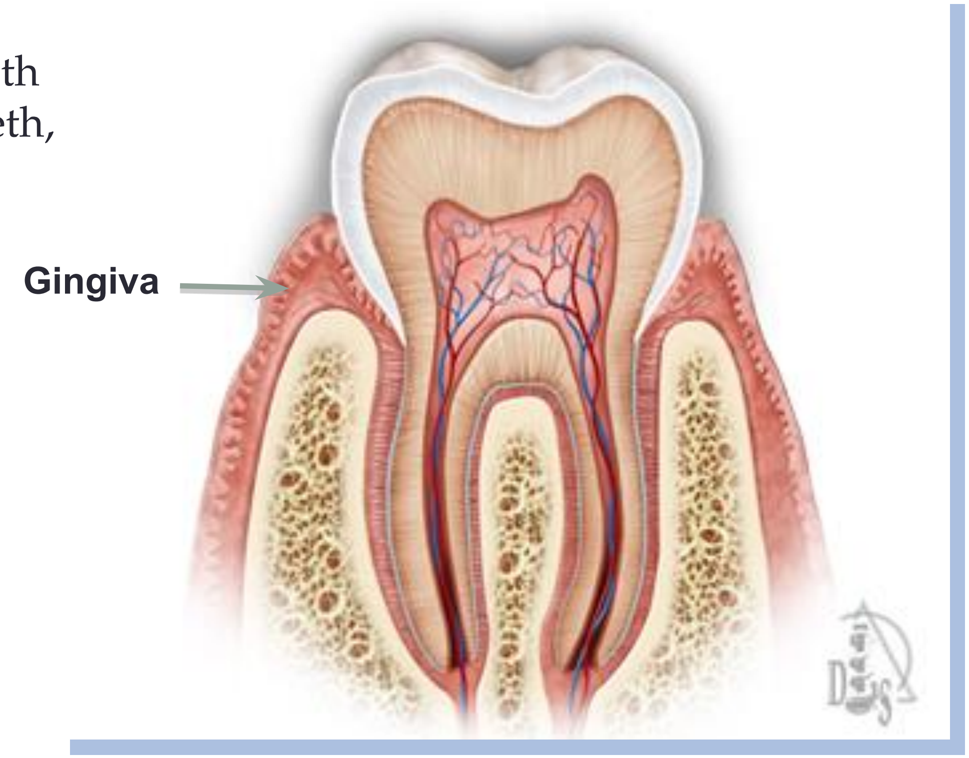
- Gingiva
- Alveolar Bone
- Periodontal Ligament
- Cementum



Dental Anatomy and Physiology

Dental Tissue—Dental Tissue[6]

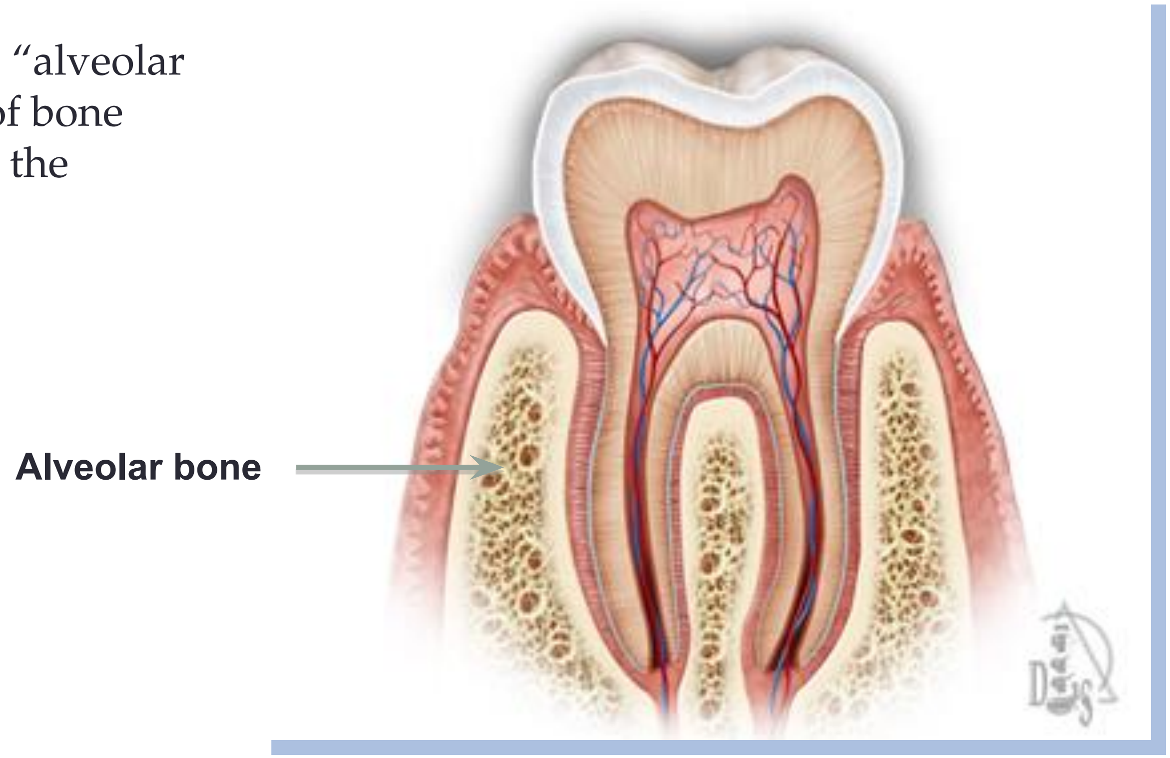
- **Gingiva:** The part of the oral mucosa overlying the crowns of unerupted teeth and encircling the necks of erupted teeth, serving as support structure for subadjacent tissues.



Dental Anatomy and Physiology

Dental Tissue—Dental Tissue[6]

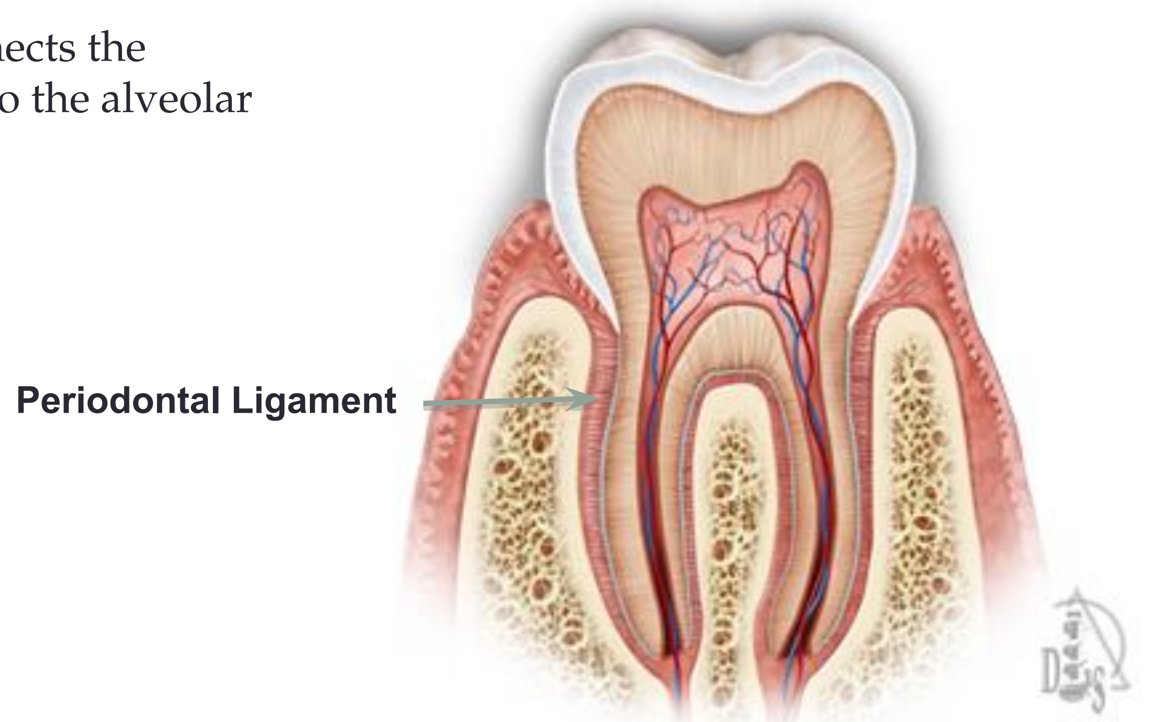
- **Alveolar Bone:** Also called the "alveolar process"; the thickened ridge of bone containing the tooth sockets in the mandible and maxilla.



Dental Anatomy and Physiology

Dental Tissue—Dental Tissue[6]

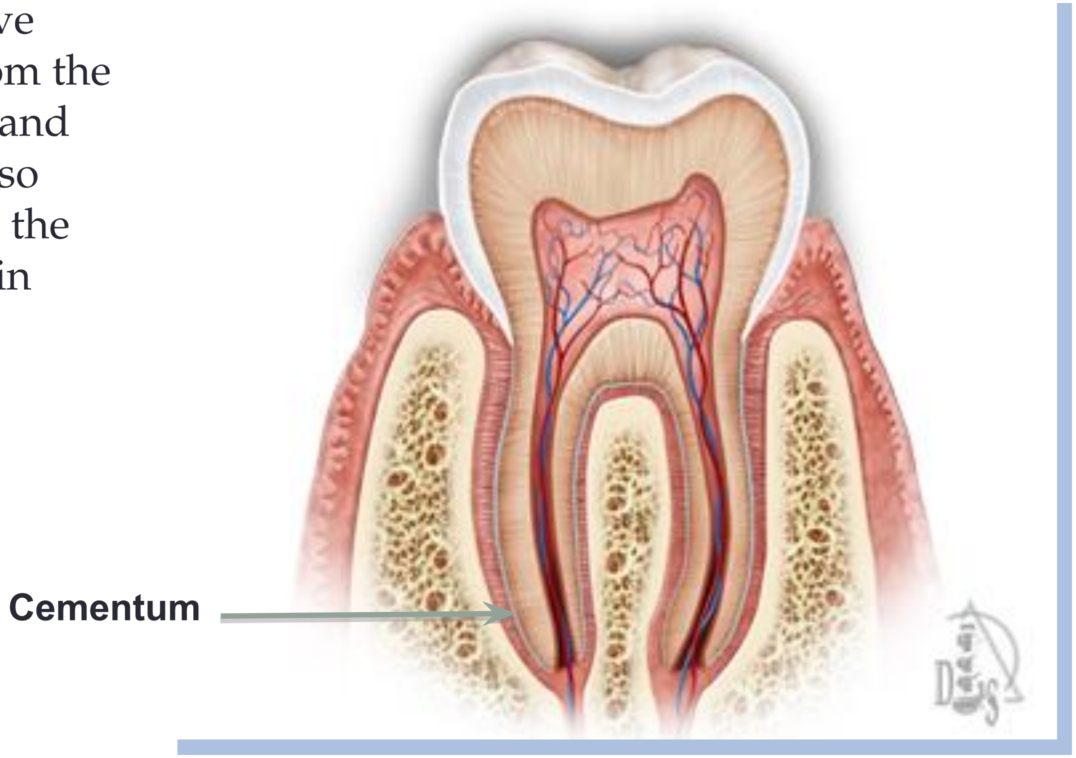
- **Periodontal Ligament:** Connects the cementum of the tooth root to the alveolar bone of the socket.



Dental Anatomy and Physiology

Dental Tissue—Dental Tissue[6]

- **Cementum:** Bonelike, rigid connective tissue covering the root of a tooth from the cementoenamel junction to the apex and lining the apex of the root canal. It also serves as an attachment structure for the periodontal ligament, thus assisting in tooth support.



(Redrawn from Fehrenbach MJ, Popowics T: *Illustrated dental embryology, histology, and anatomy*, ed 5, St Louis, 2020, Elsevier.)

Dental Anatomy and Physiology

Oral Cavity/Environment[7,8]

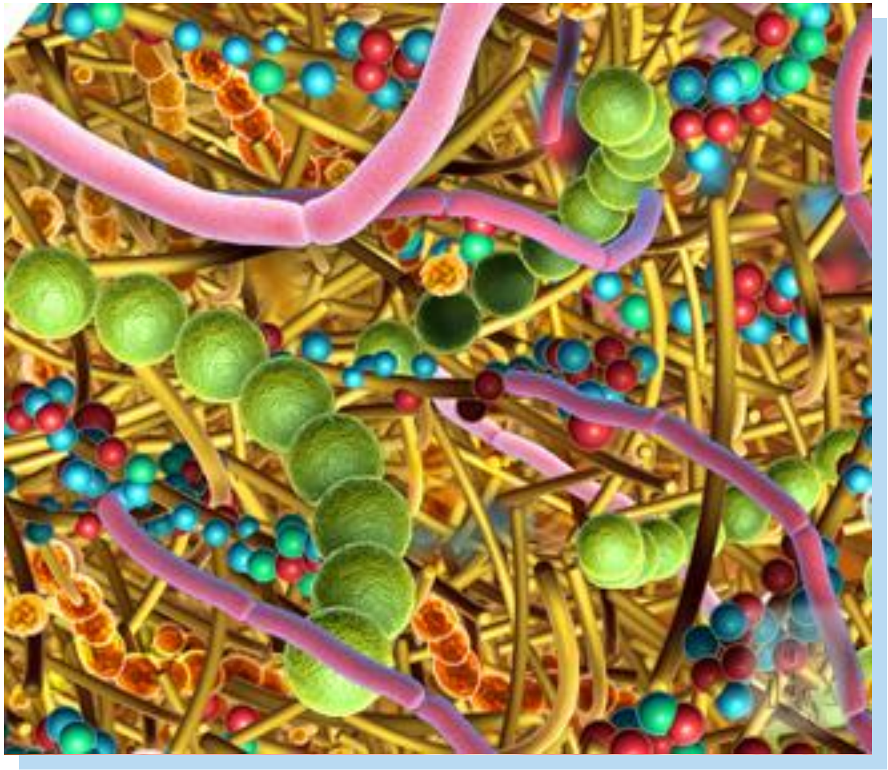
- Plaque
- Saliva
- pH Values
- Demineralization
- Remineralization

Dental Anatomy and Physiology

Oral Cavity

Plaque:[7,8]

- is a biofilm
- contains more than 600 different identified species of bacteria
- there is harmless and harmful plaque
- salivary pellicle allows the bacteria to adhere to the tooth surface, which begins the formation of plaque

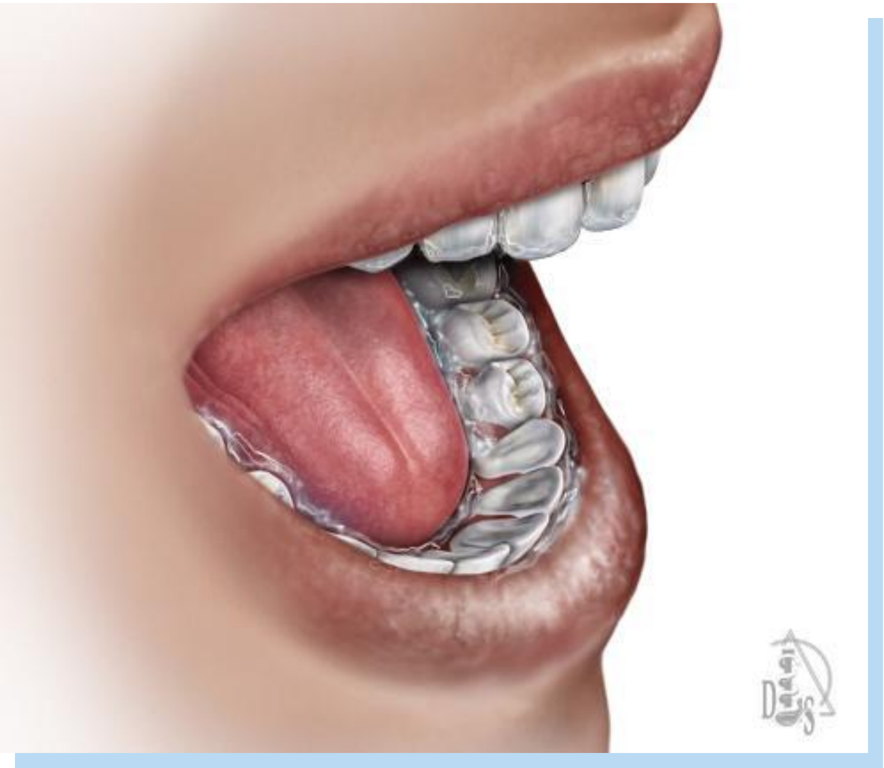


Dental Anatomy and Physiology

Oral Cavity

Saliva:[7,8]

- complex mixture of fluids
- performs protective functions:
  - lubrication—aids swallowing
  - mastication
  - key role in remineralization of enamel and dentin
  - buffering



Dental Anatomy and Physiology

Oral Cavity

pH values:[7,8]

- measure of acidity or alkalinity of a solution
- measured on a scale of 1-14
- pH of 7 indicated that the solution is neutral
- pH of the mouth is close to neutral until other factors are introduced
- pH is a factor in demineralization and remineralization

pH of Some Foods and Beverages

Food / Beverage	pH Range
Wine	2.3-3.8
Sports Drinks	2.3-4.4
Soda	2.7-3.5
Oranges	2.8-4.0
Plums	2.8-4.6
Iced Tea	3.0
Strawberries	3.0-4.2
Apple Juice	3.5
Apples	3.5-3.9
Orange Juice	3.6
Tomatoes	3.7-4.7
Vegetables	3.9-5.1
Seedless Raisins	4.0
Beer	4.0-5.0
Tea (Black)	4.2
Bananas	5.1
Natural Cheese	5.1
Whole Milk	6.7
Water	7.3

3. Strassler HE, Drisko CL, Alexander DC.

Dental Anatomy and Physiology

Oral Cavity

Demineralization:[7,8]

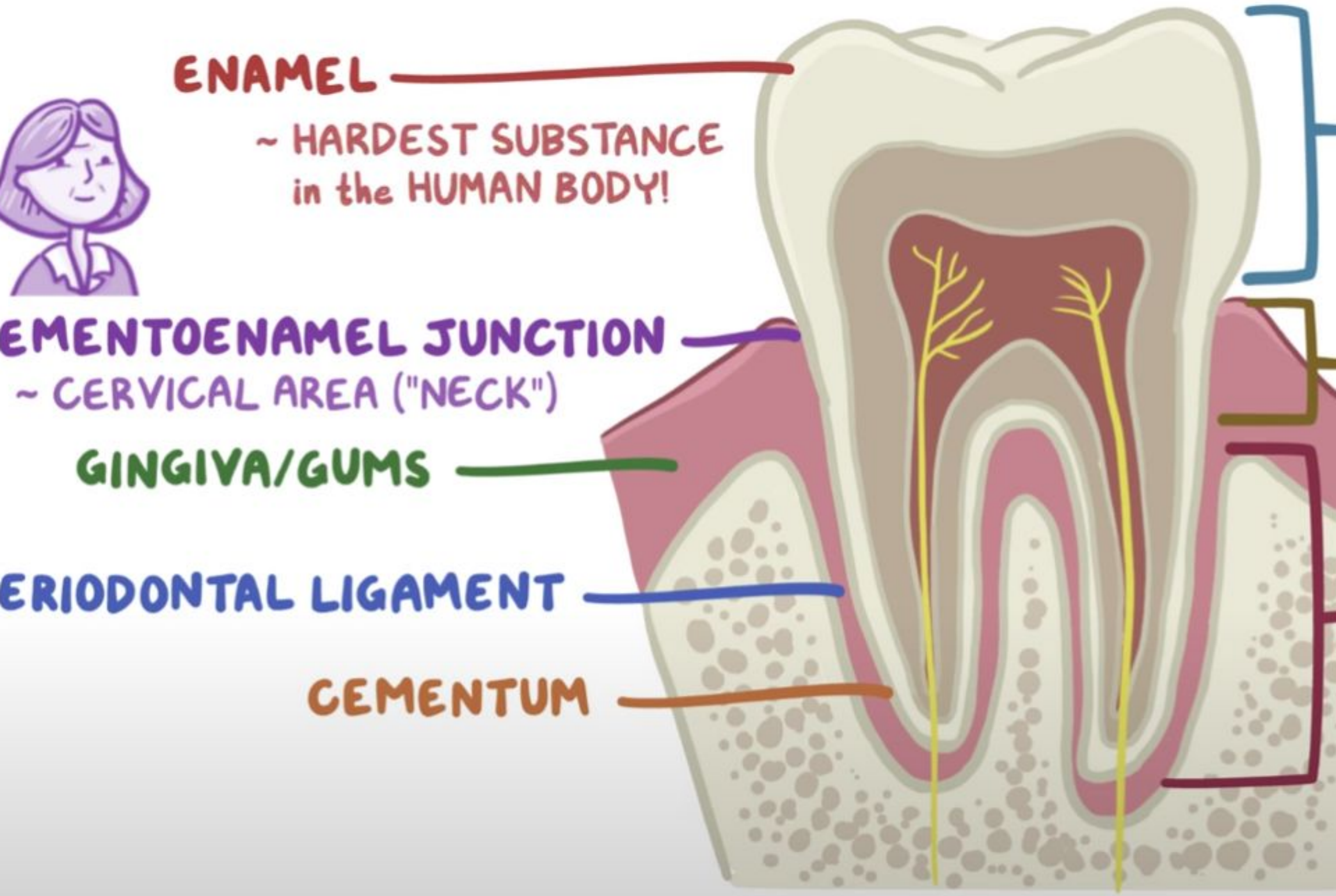
- mineral salts dissolve into the surrounding salivary fluid:
  - enamel at approximate pH of 5.5 or lower
  - dentin at approximate pH of 6.5 or lower
- erosion or caries can occur

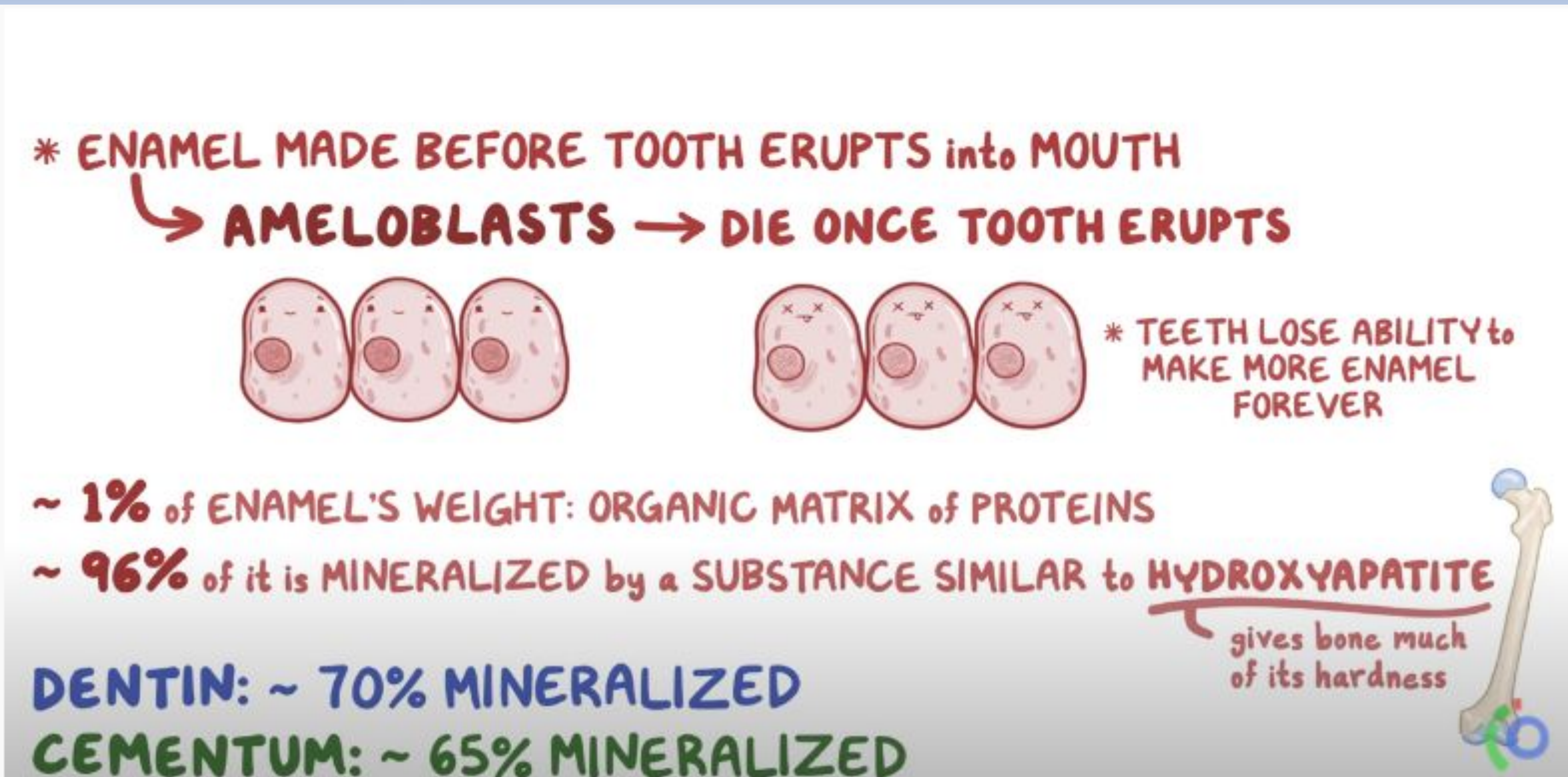
Dental Anatomy and Physiology

Oral Cavity

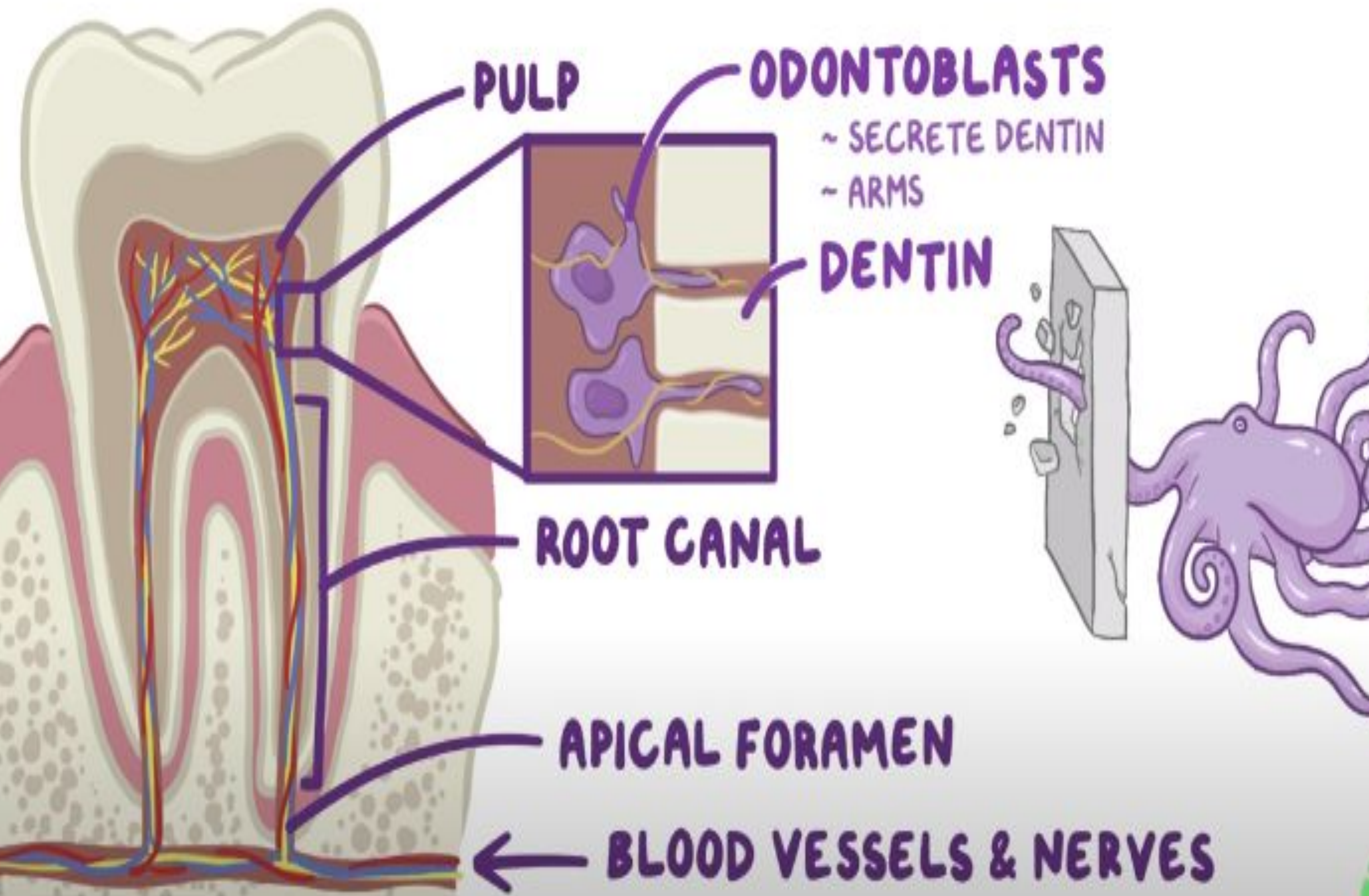
Remineralization:[7,8]

- pH comes back to neutral (7)
- saliva-rich calcium and phosphates
- minerals penetrate the damaged enamel surface and repair it:
  - enamel pH is above 5.5
  - dentin pH is above 6.5


ENAMEL
~ HARDEST SUBSTANCE in the HUMAN BODY!
EMENTOENAMEL JUNCTION
~ CERVICAL AREA ("NECK")
GINGIVA/GUMS
ERIODONTAL LIGAMENT
CEMENTUM

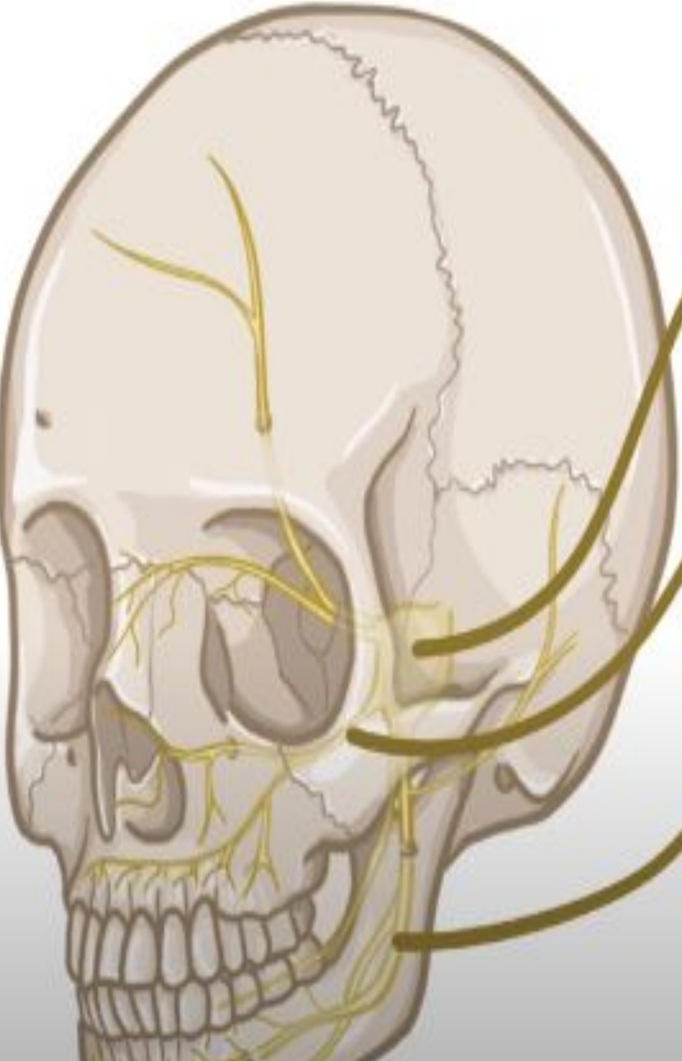

* ENAMEL MADE BEFORE TOOTH ERUPTS into MOUTH
AMELOBLASTS → DIE ONCE TOOTH ERUPTS
* TEETH LOSE ABILITY to MAKE MORE ENAMEL FOREVER
~ 1% of ENAMEL'S WEIGHT: ORGANIC MATRIX of PROTEINS
~ 96% of it is MINERALIZED by a SUBSTANCE SIMILAR to HYDROXYAPATITE
gives bone much of its hardness
DENTIN: ~ 70% MINERALIZED
CEMENTUM: ~ 65% MINERALIZED


FREE GINGIVA
GINGIVAL CREVICE / SULCUS
GINGIVAL CREVICULAR FLUID
IMMUNE PROTEINS & CELLS (e.g. neutrophils, complement proteins, & antibodies)
PROTECTS AGAINST OVERGROWTH of PATHOGENIC BACTERIA


PULP
ODONTOBLASTS
~ SECRETE DENTIN
~ ARMS
DENTIN
ROOT CANAL
APICAL FORAMEN
BLOOD VESSELS & NERVES

NERVES

└ TOUCH, PAIN, CHANGES in TEMPERATURE to the TEETH



TRIGEMINAL NERVE

(CRANIAL NERVE V)

MAXILLARY NERVE

└ INNERVATES UPPER TEETH with SMALLER BRANCHES (e.g. superior alveolar nerves)

MANDIBULAR NERVE

└ INNERVATES BOTTOM TEETH with BRANCHES (e.g. inferior alveolar nerve)


Your case is quite complicated
Why doctor ? What happened
You got a disease from the chapter which I left as optional during my studies

Enamel Erosion


Bacterial biofilms (plaque)
develop and produce acid,
which dissolves tooth enamel.
decay
enamel
dentin
pulp
dentin
plaque
infected
pulp
abscess


Enamel Cavitation
Destruction
and ADJ Spread
Bacterial Invasion
Demineralisation
Sclerotic
Tertiary Dentine

What Happens with Periodontal Disease?